

MGT 433 Managing Healthcare and Life Sciences

Winter 2026

SCHEDULE: Thursdays 6:30 – 9:20 pm Pacific Daylight Time

LOCATION: Otterson Hall, room: 3E107

INSTRUCTOR: Magnus Felke

Email: mfelke@ucsd.edu

Office hours: By appointment via Zoom or in person before or after class. Please email me for an appointment. I appreciate hearing from you and will do what I can to respond to you in a timely manner.

Teaching Assistant: Ben Whitlock

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DESCRIPTION

Health touches all of us and most of us know someone who is suffering from a potentially serious health issue. Healthcare is often described as a fundamental right. Yet, the realities of limited resources, a complex system and difficult ethics make healthcare one of the most challenging problems of our time. Every fifth dollar in the U.S. is now spent on healthcare with results that are lagging other developed nations.

This presents unique opportunities for innovation and disruption that could yield enormous benefits for patients globally. The prospect of novel business models, innovative new therapies and alternative regulatory approaches is promising and at the same time introduces new risks.

Our course will explore the business of healthcare and life sciences from a U.S. perspective and will draw comparisons and highlight alternate approaches from around the world.

You will gain an understanding of the roles, motivations, and interdependencies of key stakeholders such as patients, providers, payers, government, and industry. We will examine how these stakeholders seek to address challenges and gain a competitive advantage – both domestic and international.

During the second half of the course, our focus will shift to innovation and disruption in healthcare. You will have a chance to critically evaluate current trends and new approaches and initiatives by various players.

Upon completion of the course, you will have gained an understanding of the business side of healthcare and life sciences and career opportunities in this exciting field.

MATERIALS

Required reading consists of articles and cases. The reading assignments will be available on study.net and on BryteWave. Occasionally, I may assign additional articles, podcasts or videos.

The books *“Biodesign: The Process of Innovating Medical Technologies”* by Paul G. Yock, Stefanos Zenios, Josh Makower and *“Predictable Winners”* by Stuart E. Jackson and Ilya Trakhtenberg are meant to be optional reading and a reference for you. I encourage you to use them - especially as a resource for the innovation team project.

INSTRUCTIONAL METHODS

The course will be conducted seminar-style with some lectures and rely heavily on open discussions. Besides theoretical knowledge, teaching will rely on current case studies to provide a real-world context for discussion. Please come to class prepared and ready to discuss the assigned readings including your own thoughts on the topic.

OBJECTIVES

The course has five objectives:

1. Introduce the economic and medical challenges in healthcare and the associated societal impact.
2. Provide an understanding of the roles, motivations and challenges of the key stakeholders in healthcare.
3. Educate students on key government regulations and their implications for industry's, providers' and payers' business strategies.
4. Equip students with an understanding and appreciation for the unique challenges and trends when innovating in healthcare and life sciences.
5. Provide an understanding of career paths in healthcare and life sciences.

SCHEDULE

	Topic	Assignments:
MODULE I – Foundation in Healthcare and Life Sciences		
Week 1 January 8	Introduction to Healthcare & The Patient	Reading: “Big Med - Restaurant chains have managed to combine quality control, cost control, and innovation. Can health care?” – New Yorker article by Atul Gawande
Week 2 January 15	Providers	Reading: “How Not To Fix U.S. Health Care: Copy The Cheesecake Factory” – Forbes article by Steve Denning (rebuttal to “Big Med” article) “How to Solve The Cost Crisis in Health Care” HBR article by Robert S. Kaplan and Michael E. Porter
Week 3 January 22	Payers	Reading: “8 Basic Payment Methods in Health Care” by Kevin Quinn “U.S. Commercial Health Insurance Industry” HBS note by Susanna Gallani et al
Week 4 January 29	Pharmaceuticals & Biotechnology Guest speaker: Mark Marino, MD, Chief Medical Officer, Sunshine Bio Inc.	Reading: “Drugs, Devices, and the FDA: Part 1 – An Overview of Approval Processes for Drugs” – JACC article by Gail A. Van Norman “Value-Based Pricing for Drugs” – JAMA (June 5, 2018; Volume 319, Number 21) by Anna Kaltenboeck and Peter B. Bach Other: - Watch video about the priority review voucher: https://www.youtube.com/watch?v=ZwVC1PUvtXk
Week 5 February 5	Medical Devices & Healthcare IT	Reading: “Review of U.S. Medical Device Regulation” – Journal of Medical Devices article by Lauren Aquino Shluzas et al Other: - Healthcare business executive interview summary due.

Module II – Disruption & Innovation in Healthcare and Life Sciences		
Week 6 February 12	Mid-term Exam Private Equity in Healthcare	Reading: “Life and Debt at a Private Equity Hospital” BusinessWeek article by John Hechinger and Sabrina Willmer
Week 7 February 19	Innovation in Healthcare & Life Sciences	Reading: “Will Disruptive Innovations Cure Health Care?” – HBR article by Clayton Christensen, Richard Bohmer, and John Kenagy “Theranos: How Did a \$9 Billion Health Tech Startup End Up DOA” – Berkeley Hass Business Case by Ernesto Dal Bo Other: 5 problems for innovation project due by February 18
Week 8 February 26	Disruption in Healthcare – United States Guest speaker: Mark Johnston, Partner, ECG Management Consultants	Reading: “Oscar Health Insurance: What Lies Ahead for a Unicorn Insurance Entrant?”- Harvard Business Case “Oak Street Health: A New Model of Primary Care” – Harvard Business Case “Health Care 4.0: A vision for smart and connected health care” by Jingshan Li & Pascale Carayon
Week 9 March 5	Disruption in Healthcare – International Guest speaker: Kevin Pegels, Chief Operations Officer at Illumina	Reading: “Health City Cayman Islands” – Harvard Business Case “Cleveland Clinic Abu Dhabi (Abridged)” – Harvard Business Case (optional) Other: - Watch video about Narayana Health: https://www.youtube.com/watch?v=wXvTzpapa-s
Week 10 March 12	Project Presentations & Discussion	Reading: “Vital Directions and National Will” – JAMA editorial by Donald J. Berwick, MD “America’s Sickness Economy” – Financial Times article by Tej Parikh (optional) Other: - Innovation project presentation. - Please fill out the course evaluation.

ASSIGNMENTS:

Reading Assignments:

Please read the assigned readings prior to the respective class sessions. This will allow you to come prepared with your own thoughts, examples and questions and allow you to participate fully in class discussions.

All readings can be found on Canvas under “Study.net Materials”. You can download the entire reader or select individual articles.

Class Participation and Professionalism:

Being present and active participation are key in this class. Much of your learning will happen during class discussions through the exchange of ideas. This can mean asking questions to enhance your (and others’) understanding of the material as well as contributing with examples and thoughts that demonstrate your understanding of what we discuss. The expectation is that you come prepared with comments and/or questions that are relevant to the class. You may also consider bringing news articles and stories that pertain to the topic of the class session.

Professionalism is expected. That means treating others with courtesy, respect, and listening to their points of view as well as working collaboratively in teams. It also means being *present* during class and refraining from the use of laptops, tablets and phones unless required. Lastly, it is expected that you arrive on time and remain until the end of each class meeting.

Healthcare Business Executive Interview:

There will be one “smaller” team project that you will do in teams of two. You will reach out to a professional who works in healthcare or life sciences in a business capacity at a senior level (director or above) and interview them. You could use LinkedIn to reach out to Rady or UCSD alumni. The goal is to understand a) what they see as key trends, business challenges and opportunities in their area of healthcare and/or life sciences, b) how they got to where they are and what career advice they may have for you. You will submit a 1.5-2 page (single-spaced) written summary of what you learned from the interview. This should not be a transcript of your conversation but a synopsis of what you discussed and the key insights that you took away from the conversation.

This assignment is due by February 5.

Healthcare Innovation Team Project:

There will be one “big” team project including a final team presentation in lieu of a final exam. The project allows you to apply what you have learned about healthcare stakeholders, incentives, government regulation, and the challenges in healthcare. The assignment is for your team to develop an innovation idea in healthcare or life sciences and demonstrate how it is advantageous from a business perspective. As a first step you may want to brainstorm problems and choose one to focus on as a team. This may involve conducting research on the size of the problem, which stakeholders it impacts, what their motivations are and whether solutions are available. From there, you will propose one or more solutions as your innovation idea. This should include a projection of demand and a financial analysis of the opportunity and consider reimbursement. Whenever possible, data should be used

to support your case. Please also consider risks and opportunities for your proposal. Your concept needs to be unique and something that does not already exist. Your idea can be U.S. focused, global or focused on a particular country other than the U.S. Technological feasibility is important but secondary to the business considerations. (Please do not spend too much time exploring technical options.) Both for-profit and non-profit concepts are feasible as long as they are sustainable.

The final output of this project will be a team presentation where you will share your idea with the class. Presentations will be evaluated on how well thought-out your concept is, completeness, application of learned concepts, the level of preparation and effectiveness of delivery as well as the attractiveness and probability of success of your proposed innovation.

I will ask each of you to submit a "Peer Evaluation Form" at the end of the quarter. The form will ask you to evaluate the contributions of your group members based on specific criteria, such as leadership, communication, and teamwork. The evaluations will be kept confidential and will not be shared with other group members. The evaluations will be used to inform my assessment of individual contributions to group work and ensure a fair grade for each group member. If the majority of your group members say that someone in the group contributed significantly less, I will decrease that person's grade. Submitting the peer evaluation form is required.

I will provide more detailed information when I introduce the assignment in class during week 6.

- *February 18 (BEFORE class meeting #7): Submit the 5 or more problems to solve that you have identified.*
- *March 12 (class meeting #10): Final team presentations, 10 minutes*

Mid-Term Exam:

The mid-term exam will consist of multiple-choice and short-answer/essay questions to assess your knowledge of the concepts around healthcare stakeholders in Module I (weeks 1-5). The exam will cover any class material up to that point including readings, lectures and discussions.

Reflection Memos:

You will write reflection memos on the four articles highlighted in **red** on the course schedule. These reflections should be ~ 1 ½ pages in length and not exceed 2 pages in length (single-spaced). These submissions are **due on the day the respective article will be discussed in class**.

In the reflection memo, please discuss your thoughts on the article e.g. what surprised you? How does it relate to what we have discussed in class? Do you have personal experiences that you can relate to the article? What thoughts do you have that expand on the article e.g. based on current events or other articles? What do you think will happen next and why? The key point to remember is that the paper is not a test. The goal is not to summarize the article or to answer an exam question, but rather it is to explore your own thoughts and ideas about a topic. The following reference may help you structure your reflection memo:

<https://coffin.files.wordpress.com/2012/09/writing-an-article-reflection.pdf>

GRADING:

Assignments	Points	Percent of Grade
Class Participation & Professionalism	200	20%
Healthcare Executive Interview	100	10%
Healthcare Innovation Team Project	250	25%
Reflection Memos	200	20%
(Mid-Term) Exam	250	25%
Total	1000	100%

COURSE POLICIES

The majority of content and learning will be covered during class sessions in the form of lectures and especially discussions. Therefore, you will take away more from the course if you attend the class meetings.

During the class, you may want to share healthcare examples that are personal or relate to people you know. Please use your judgement in what you are comfortable sharing and respect others' privacy (e.g. by keeping their identity confidential).

UCSD email and Canvas will be used as the primary means of communication and material delivery within this course. Be sure to have consistent access and regularly check both.

While AI tools will likely not be as helpful in this course, the use of AI tools is allowed. **However, AI tools must be noted as a source when used for assignments.** It should be noted that the accuracy of large language models is often poor and that they do not serve as a replacement for human understanding. It is the responsibility of each student to ensure their use of content generated by AI tools does not constitute academic misconduct, such as plagiarism (intentional or unintentional) or cheating.

ACADEMIC INTEGRITY

Academic Integrity is expected of everyone at UC San Diego. This means that you must be honest, fair, responsible, respectful, and trustworthy in all of your actions. Lying, cheating or any other forms of dishonesty will not be tolerated because they undermine learning and the University's ability to certify students' knowledge and abilities. Thus, any attempt to get, or help another get, a grade by cheating, lying or dishonesty will be reported to the Academic Integrity Office and will result in sanctions. Sanctions can include an F in this class and suspension or dismissal from the University.

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

You can learn more about academic integrity at <https://academicintegrity.ucsd.edu/>

The complete UCSD Policy on Integrity of Scholarship can be viewed at:

<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the instructor and OSD liaison.

Students requesting accommodations for this course due to a disability must provide a current Authorization for Accommodation (AFA) letter (paper or electronic) issued by the Office for Students with Disabilities (<https://osd.ucsd.edu/portal/tutorial.html>) Students are required to discuss accommodation arrangements with instructors and OSD liaisons in the department **in advance** of any exams or assignments. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.

FINANCIAL AID

Satisfactory Academic Progress (SAP) refers to the academic standards students must maintain to remain eligible for federal, state, and institutional financial aid. If you are receiving financial aid, please ensure you review the SAP requirements and the appeals process. (<https://fas.ucsd.edu/forms-and-resources/sap/index.html>)

NONDISCRIMINATION POLICY STATEMENT

The University of California, in accordance with applicable Federal and State law and University policy, does not discriminate on the basis of race, color, national origin, religion, sex, gender identity, pregnancy, physical or mental disability, medical condition (cancer related or genetic characteristics), ancestry, marital status, age, sexual orientation, citizenship, or service in the uniformed services. The University also prohibits sexual harassment. This nondiscrimination policy covers admission, access, and treatment in University programs and activities.

TITLE IX

The Office for the Prevention of Harassment & Discrimination (OPHD) provides assistance to students, faculty, and staff regarding reports of bias, harassment, and discrimination. OPHD is the UC San Diego Title IX office. Title IX of the Education Amendments of 1972 is the federal law that prohibits sex discrimination in educational institutions that are recipients of federal funds. Rady students have the right to an educational environment that is free from harassment and discrimination.

You can make a complaint of harassment or discrimination — or simply make an appointment to find out more information — by contacting OPHD.

- (858) 534-8298
- ophd@ucsd.edu
- [Overview for Students](#) webpage

Students may feel more comfortable discussing their particular concern with a trusted employee. This may be a Rady student affairs staff member, a department Chair, a faculty member or other University official. These

individuals have an obligation to report incidents of sexual violence and sexual harassment to OPHD. This does not necessarily mean that a formal complaint will be filed.

If you find yourself in an uncomfortable situation, ask for help. The Rady School of Management is committed to upholding University policies regarding nondiscrimination, sexual violence and sexual harassment.